MITHUN CHAKRABORTY

Assistant Research Scientist	Office: 2709 Beyster
Computer Science and Engineering (CSE) Division	
Electrical Engineering & Computer Science (EECS)	Mailing address: 2260 Hayward St.
University of Michigan Ann Arbor	Ann Arbor, MI 48109
https://sites.google.com/umich.edu/mithunchakra/	E-mail: *dcsmc@umich.edu*

RESEARCH INTERESTS

Artificial Intelligence: Multi-Agent Systems; Agent-Based Models; Algorithmic Game Theory; Computational Economics and Finance: Fair Allocation, Collective Forecasting (Prediction Markets), Financial Market Simulation

EDUCATION

Ph.D. in Computer Science May 2017
Washington University in St. Louis (WUSTL)
Advisor: Sanmay Das (now at George Mason University)
Dissertation topic: **On the Aggregation of Subjective Inputs from Multiple Sources**

Bachelor of Electronics & Telecommunication Engineering (Hons.) Jun 2009
Jadavpur University, Kolkata, India

SELECTED AWARDS AND RECOGNITIONS

ICAIF24 Best Paper Award 2024

Department Chair Award for Outstanding Teaching 2016
Conferred by the Department of Computer Science and Engineering, WUSTL

WU-CIRTL Practitioner Level Recognition 2016
Conferred by the Teaching Center, WUSTL

PROFESSIONAL EXPERIENCE

Assistant Research Scientist Feb 2020 – present
Faculty member, Strategic Reasoning Group, AI Lab,
EECS, University of Michigan Ann Arbor;
Affiliate Member, MIDAS

LEO Intermittent Lecturer Fall 2023, Winter 2023, Winter 2022
CSE, University of Michigan Ann Arbor

Post-doctoral Research Fellow Jun 2017 – Jan 2020
Host: Yair Zick (now at University of Massachusetts, Amherst),

Data-Driven Strategic Collaboration Group,
Department of Computer Science, National University of Singapore

Graduate Research Assistant

Washington University in St. Louis — Fall 2013 – Spring 2017
Virginia Tech — Fall 2012 – Spring 2013
Rensselaer Polytechnic Institute (RPI) — Fall 2009 – Spring 2012

FUNDING

- **Cooperative AI Foundation Research Grant** ($347,424). Title: *Measuring cooperation among competing AI algorithms* (1 Nov 2024 – 31 Oct 2026). **Role**: Co-Principal Investigator.
- **NSF CRII Award** #2153184 ($175,000). Title: *CRII: RI: Analysis and Applications of Multi-Level Games* (1 May 2022 – 30 Apr 2025). **Role**: Principal Investigator.
- **Facebook (now Meta) Research Award** in response to the request for proposals on agent-based user interaction simulation ($50,000). Title: *Empirical game-theoretic analysis for web-enabled simulation* (2020). **Role**: Co-author and senior personnel.

WORKING PAPERS

[W1] Christine Konicki, **Mithun Chakraborty**, Michael Wellman. *Computing Perfect Bayesian Equilibria with Application to Strategy Exploration in Extensive-Form Empirical Games*.

[W2] Nicholas Schumacher, **Mithun Chakraborty**, Sindhu Kutty. *Wealth vs. Wisdom: How budgets limit information aggregation in prediction markets*.

[W3] Noah Lincke, **Mithun Chakraborty**, Sindhu Kutty. *A Comparative Study of Elicitation Granularity in Prediction Markets Under Trader Misinformation*.

JOURNAL ARTICLES

[J1] Feiran Jia, Aditya Mate, Zun Li, Shahin Jabbari, **Mithun Chakraborty**, Milind Tambe, Michael Wellman, Yevgeniy Vorobeychik. *A Game-Theoretic Approach for Hierarchical Epidemic Control*. Autonomous Agents and Multiagent Systems, Vol. 39, Article No. 14, pp. 1–37, February 2025.

[J2] **Mithun Chakraborty**, Erel Segal-Halevi, Warut Suksompong. *Weighted Fairness Notions for Indivisible Items Revisited*. ACM Transactions on Economics and Computation (TEAC), Vol. 12 No. 3, Article No. 9, pp. 1–45, September 2024. Supersedes [C6].

[J3] Nawal Benabbou, **Mithun Chakraborty**, Ayumi Igarashi, Yair Zick. *Finding Fair and Efficient Allocations under Matroid Rank Valuations*. ACM Transactions on Economics and Computation (TEAC), Vol. 9 No. 4, Article No. 21, pp. 1–41, December 2021. Supersedes [C9].

[J4] **Mithun Chakraborty**, Ulrike Schmidt-Kraepelin, Warut Suksompong. *Picking Sequences and Monotonicity in Weighted Fair Division*. Journal of Artificial Intelligence (AIJ). Vol. 301, Article 103578, pp. 1–21, December 2021. Supersedes [C8].

[J5] **Mithun Chakraborty**, Ayumi Igarashi, Warut Suksompong, Yair Zick. *Weighted Envy-Freeness in Indivisible Item Allocation*. ACM Transactions on Economics and Computation (TEAC), Vol. 9 No. 3, Article No. 18, pp. 1–39, September 2021. Supersedes [C10].

[J6] Nawal Benabbou, **Mithun Chakraborty**, Xuan-Vinh Ho, Jakub Sliwinski, Yair Zick. *The Price of Quota-based Diversity in Assignment Problems*. ACM Transactions on Economics and Computation (TEAC) Vol. 8 No. 3, Article No. 14, pp. 1–32, September 2020. Supersedes [C12].

[J7] Nawal Benabbou, **Mithun Chakraborty**, Yair Zick. *Fairness and Diversity in Public Resource Allocation Problems*. Bulletin of the Technical Committee on Data Engineering Vol. 42 No. 3, Special Issue on Fairness, Diversity, and Transparency in Data Systems, pp. 64–75, September 2019. Invited paper.

[J8] Pabitra Pal Choudhury, Sudhakar Sahoo, **Mithun Chakraborty**, Subir Bhandari, Amita Pal. *Investigation of the global dynamics of cellular automata using Boolean derivatives*. Computers and Mathematics with Applications (Elsevier), Vol. 57 Issue 8, pp. 1337–1351, 2009.

[J9] Deepyaman Maiti, **Mithun Chakraborty**, Ayan Acharya, Amit Konar. *A partly deterministic and partly stochastic scheme for the identification of fractional-order processes*. International Journal of Advanced Intelligence Paradigms (Inderscience), Vol. 1 No. 3, pp. 332–357, 2009.

ARCHIVAL CONFERENCE PAPERS (PEER-REVIEWED)

[C1] Christine Konicki, **Mithun Chakraborty**, Michael Wellman. *Policy Abstraction and Nash Refinement in Tree-Exploiting PSRO*. Accepted to 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2025 [full paper: 24.5% acceptance rate].

[C2] Chris Mascioli, Anri Gu, Yongzhao Wang, **Mithun Chakraborty**, Michael Wellman. *A Financial Market Simulation Environment for Trading Agents Using Deep Reinforcement Learning*. Proc. of 5th ACM International Conference on AI in Finance (ICAIF), pp. 117–125, 2024 [oral presentation]. **ICAIF24 Best Paper Award**.

[C3] Anri Gu, Yongzhao Wang, Chris Mascioli, **Mithun Chakraborty**, Rahul Savani, Ted Turocy, Michael Wellman. *The Effect of Liquidity on the Spoofability of Financial Markets*. Proc. of 5th ACM International Conference on AI in Finance (ICAIF), pp. 239–247, 2024 [oral presentation].

[C4] Christine Konicki, **Mithun Chakraborty**, Michael Wellman. *Exploiting Extensive-Form Structure in Empirical Game-Theoretic Analysis*. Proc. of 18th Conference on Web and Internet Economics (WINE), pp. 132–149, 2022 [30.2% acceptance rate].

[C5] Zun Li, Feiran Jia, Aditya Mate, Shahin Jabbari, **Mithun Chakraborty**, Milind Tambe, Yevgeniy Vorobeychik. *Solving Structured Hierarchical Games Using Differential Backward Induction*. Proc. of Machine Learning Research (PMLR), Vol. 180: 38th Conference on

Uncertainty in Artificial Intelligence (UAI), pp. 1107–1117, 2022 [32.3% acceptance rate]. **Selected for oral presentation** [5% of submissions].

[C6] **Mithun Chakraborty**, Erel Segal-Halevi, Warut Suksompong. *Weighted Fairness Notions for Indivisible Items Revisited*. Proc. of 36th AAAI Conference on Artificial Intelligence (AAAI), pp. 4949–4956, 2022 [15% acceptance rate].

[C7] Blake Martin, **Mithun Chakraborty**, Sindhu Kutty. *Timing is Money: The Impact of Arrival Order in Beta-Bernoulli Prediction Markets*. Proc. of 2nd ACM International Conference on AI in Finance (ICAIF), Article No.: 41, pp. 1–9, 2021 [49% acceptance rate].

[C8] **Mithun Chakraborty**, Ulrike Schmidt-Kraepelin, Warut Suksompong. *Picking Sequences and Monotonicity in Weighted Fair Division*. Proc. of 30th International Joint Conference on Artificial Intelligence (IJCAI), pp. 82–88, 2021 [13.9% acceptance rate].

[C9] Nawal Benabbou, **Mithun Chakraborty**, Ayumi Igarashi, Yair Zick. *Finding Fair and Efficient Allocations when Valuations Don't Add Up*. Proc. of 13th Symposium on Algorithmic Game Theory (SAGT), pp. 32–46, 2020.

[C10] **Mithun Chakraborty**, Ayumi Igarashi, Warut Suksompong, Yair Zick. *Weighted Envy-Freeness in Indivisible Item Allocation*. Proc. of 19th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS), pp. 231–239, 2020 [23% acceptance rate].

[C11] Nawal Benabbou, **Mithun Chakraborty**, Edith Elkind, Yair Zick. *Fairness Towards Groups of Agents in the Allocation of Indivisible Items*. Proc. of 28th International Joint Conference on Artificial Intelligence (IJCAI), pp. 95–101, 2019 [17.9% acceptance rate].

[C12] Nawal Benabbou, **Mithun Chakraborty**, Xuan-Vinh Ho, Jakub Sliwinski, Yair Zick. *Diversity Constraints in Public Housing Allocation*. Proc. of 17th International Conference on Autonomous Agents and Multiagent Systems (AAMAS), pp. 973–981, 2018 [Oral presentation: 25.3% acceptance rate].

[C13] **Mithun Chakraborty**, Kai Yee Phoebe Chua, Sanmay Das, Brendan Juba. *Coordinated Versus Decentralized Exploration In Multi-Agent Multi-Armed Bandits*. Proc. of 26th International Joint Conference on Artificial Intelligence (IJCAI), pp. 164–170, 2017 [26.0% acceptance rate].

[C14] **Mithun Chakraborty**, Sanmay Das. *Trading On A Rigged Game: Outcome Manipulation In Prediction Markets*. Proc. of 25th International Joint Conference on Artificial Intelligence (IJCAI), pp. 158–164, 2016 [< 25% acceptance rate].

[C15] **Mithun Chakraborty**, Sanmay Das. *Market Scoring Rules Act As Opinion Pools For Risk-Averse Agents*. Advances in Neural Information Processing Systems (NeurIPS), pp. 2350–2358, 2015. **Selected for spotlight presentation [16.6% of accepted papers; 3.6% of submissions]**.

[C16] **Mithun Chakraborty**, Sanmay Das, Justin Peabody. *Price Evolution in a Continuous Double Auction Prediction Market With a Scoring-Rule Based Market Maker*. Proc. of 29th AAAI Conference on Artificial Intelligence (AAAI), pp. 835–841, 2015 [26.7% acceptance rate].

[C17] **Mithun Chakraborty**, Sanmay Das, Allen Lavoie, Malik Magdon-Ismail, Yonatan Naamad. *Instructor Rating Markets*. Proc. of 27th AAAI Conference on Artificial Intelligence (AAAI), pp. 159–165, 2013 [29% acceptance rate].

[C18] Aseem Brahma, **Mithun Chakraborty**, Sanmay Das, Allen Lavoie, Malik Magdon-Ismail. *A Bayesian Market Maker*. Proc. of 13th ACM Conference on Electronic Commerce (EC), pp. 215–232, 2012 [33.3% acceptance rate].

[C19] **Mithun Chakraborty**, Sanmay Das, Malik Magdon-Ismail. *Near-Optimal Target Learning With Stochastic Binary Signals*. Proc. of 27th Conference on Uncertainty in Artificial Intelligence (UAI), pp. 69–76, 2011 [33.7% acceptance rate].

NON-ARCHIVAL WORKSHOP AND CONFERENCE PAPERS AND PRESENTATIONS

[N1] Noah Lincke, **Mithun Chakraborty**, Sindhu Kutty. *A Comparative Study of Elicitation Granularity in Prediction Markets Under Trader Misinformation*.
- 5th Games, Agents, and Incentives Workshop (GAIW), 2023.
- 3rd International Workshop on Modelling Uncertainty in the Financial World (MUFin'23), 2023.

[N2] Zun Li, Feiran Jia, Aditya Mate, Shahin Jabbari, **Mithun Chakraborty**, Milind Tambe, Yevgeniy Vorobeychik. *Solving Structured Hierarchical Games Using Differential Backward Induction*.
- ICLR Workshop on Gamification and Multiagent Solutions, 2022.

[N3] Blake Martin, **Mithun Chakraborty**, Sindhu Kutty. *Timing is Money: The Impact of Arrival Order in Beta-Bernoulli Prediction Markets*.
- NeurIPS Workshops on Learning in Presence of Strategic Behavior, and Learning and Decision-Making with Strategic Feedback, 2021 (jointly accepted).
- Preliminary version presented as student poster at 8th International Workshop on Computational Social Choice (COMSOC), 2021.

[N4] Feiran Jia, Aditya Mate, Zun Li, Shahin Jabbari, **Mithun Chakraborty**, Milind Tambe, Michael Wellman, Yevgeniy Vorobeychik. *A Game-Theoretic Approach for Hierarchical Policy-making*.
- 2nd International Workshop on Autonomous Agents for Social Good (AASG), 2021.

[N5] **Mithun Chakraborty**, Ulrike Schmidt-Kraepelin, Warut Suksompong. *Picking Sequences and Monotonicity in Weighted Fair Division*.
- 3rd Games, Agents, and Incentives Workshop (GAIW), 2021.

[N6] **Mithun Chakraborty**, Ayumi Igarashi, Warut Suksompong, Yair Zick. *Weighted Envy-Freeness in the Allocation of Indivisible Goods*.
- 2nd Games, Agents, and Incentives Workshop (GAIW@AAMAS), 2020.

[N7] Nawal Benabbou, **Mithun Chakraborty**, Ayumi Igarashi, Yair Zick. *Finding Fair and Efficient Allocations when Valuations Don't Add Up*.
- AI for Social Good, Harvard CRCS Workshop, 2020.
- 2nd Games, Agents, and Incentives Workshop (GAIW@AAMAS), 2020.

[N8] Nawal Benabbou, **Mithun Chakraborty**, Edith Elkind, Yair Zick. *Fairness Towards Groups of Agents in the Allocation of Indivisible Items*.

- 21st Annual Conference of the French Society for Operational Research and Decision Support (ROADEF), 2020, under the title *Partage Équitable de Ressources á des Groupes d'Agents*. Abstract only.
- AAMAS workshop on Fair Allocation in Multiagent Systems (FAMAS), 2019.

[N9] Nawal Benabbou, **Mithun Chakraborty**, Xuan-Vinh Ho, Jakub Sliwinski, Yair Zick. *The Assignment Problem with Diversity Constraints with an application to Ethnic Integration in Public Housing.*

- 20th Annual Conference of the French Society for Operational Research and Decision Aiding (ROADEF), 2019, under the title *Un Problème d'Affectation avec des Contraintes de Diversité: Complexité et Prix de la Diversité*. Abstract only.
- 7th International Workshop on Computational Social Choice (COMSOC), 2018.

[N10] **Mithun Chakraborty**, Kai Yee Phoebe Chua, Sanmay Das, Brendan Juba. *Coordinated Versus Decentralized Exploration In Multi-Agent Multi-Armed Bandits*.

- 8th International Workshop on Cooperative Games and Multiagent Systems (CoopMAS) at AAMAS, 2017.
- 2nd Workshop on Learning, Inference and Control of Multi-Agent Systems (MALIC) at NeurIPS, 2016.

[N11] **Mithun Chakraborty**, Sanmay Das. *On Manipulation in Prediction Markets When Participants Influence Outcomes Directly*.

- 4th Workshop on Social Computing and User Generated Content (SCUGC) at EC, 2014.
- Collective Intelligence Conference (CI), 2014. Abstract only.

[N12] **Mithun Chakraborty**, Sanmay Das, Allen Lavoie, Malik Magdon-Ismail, Yonatan Naamad. *Instructor Rating Markets*.

- 1st Workshop on Social Computing and User Generated Content (SCUGC) at EC, 2011.
- 2nd Conference on Auctions, Market Mechanisms, and Their Applications (AMMA), 2011. Abstract only.

PROFESSIONAL SERVICE

Local Organizing Committee Co-chair: AAMAS 2025 (Detroit, MI, USA)

Senior Programming Committee Member: IJCAI (2021)

Programming Committee Board Member: IJCAI (2022-2024)

Programming Committee Member: AAAI (2022, 2021, 2020, 2019, 2018, 2015, 2014); IJCAI (2025, 2019); IJCAI-PRICAI (2020); IJCAI-ECAI (2018); AAMAS (2023, 2022, 2021, 2020, 2019); AIES (2024, 2023); EC (2022, 2018); DAI (2022, 2021); GAIW (2021, 2020); FAMAS (2019); CoopMAS (2017)

Journal Reviewer: SIAM Journal of Computing (2024); Management Science (2023, 2022); Information Processing Letters (2022); Artificial Intelligence Journal (2024, 2023, 2022); Autonomous Agents and Multi-Agent Systems (2022, 2021); Journal of Artificial Intelligence Research (2019); Frontiers of Computer Science, Springer (2014)

Conference Reviewing / Sub-Reviewing: UAI (2025); GAIW (2025); EAAMO (2022); SODA (2022); EC (2021, 2020); ACC (2022, 2021); WINE (2019, 2017); AIES (2019); AAMAS (2018); NetEcon (2017); ECAI (2014)

Research Proposal Reviewer: Israel Science Foundation (2023)

TEACHING

Instructor: EECS 592 Foundations of AI, UMich — Fall 2023, Winter 2023, Winter 2022

Guest lecturer:
– EECS 592 Foundations of AI, UMich — Fall 2022
– EECS 498-006/007 Machine Learning Research Experience, UMich — Fall 2024

Co-instructor: CS4261/CS5461 Algorithmic Mechanism Design, NUS — Semester 1, 2019/2020

Instructor: CSE 316A Social Network Analysis, WUSTL — Fall 2015

Teaching Assistant:
– CSE 516A Multi-Agent Systems (instructor: Sanmay Das), WUSTL — Spring 2014
– CSCI 2500 Computer Organization (instructor: Chris Carothers), RPI — Fall 2009

DISSERTATION COMMITTEE MEMBERSHIP

- Christine Konicki, CSE, University of Michigan — 2024
- Cristian-Paul Bara, CSE, University of Michigan — 2022

RESEARCH MENTORSHIP AND EVALUATION

- **Undergraduate students mentored/co-mentored**
  - Blake Martin (jointly with Sindhu Kutty)
    EECS 499 (directed study): Learning Generative Models with Prediction Markets (Winter 2021)
  - Noah Lincke (jointly with Sindhu Kutty)
    SURE 2021 CSE Project #10: Does Wealth Matter? Learning Generative Models with Prediction Markets (Summer 2021)
    EECS 499 (directed study): Multi-agent simulations of Beta-Bernoulli prediction markets (Fall 2021; Winter, Summer 2022)

- Hui Zhi (jointly with Sindhu Kutty)
  EECS 499 (directed study): A Comparative Study on Prediction Market Design (Fall 2021)
- Dhanya Narayanan (jointly with Sindhu Kutty)
  UROP + ExploreCSR: A Probabilistic Approach to Market Design for Collective Forecasting (Fall 2021; Winter 2022)
- Jasmine Tan (jointly with Michael Wellman)
  Project: Extending financial market simulator MarketSim to aid the study of pathology propagation on a market network (Summer, Fall 2022)
- James Edwards (jointly with Sindhu Kutty)
  EECS 499 (directed study): Studying Gaussian prediction markets with budget-limited traders (Winter 2023)
  BS in Data Science thesis (high honors): The Simulation and Analysis of Gaussian Prediction Market (Fall 2024)
- Cohort: Alex Jian, Zhaoyuan Liang, Xuteng Luo, Bohan Shu, Norman Wen, Yolanda Zhou (jointly with Michael Wellman)
  Project: Python reimplementation of financial market simulator MarketSim (Winter 2023 – Fall 2024)
- Riya Chakravarty (jointly with Sindhu Kutty)
  ExploreCSR + EECS 399 (directed study): Cost function prediction markets (Winter 2024)
  EECS 499 (directed study): Decentralized Exchange Systems: Uniswap (Fall 2024)
- Rishith Seelam (jointly with Michael Wellman)
  SURE 2024 CSE Project #27: Replication experiments on market inefficiencies under diverse trading activities using agent-based financial market simulator PyMarketSim (Summer 2024)
- Nicholas Schumacher (jointly with Sindhu Kutty)
  EECS 399 + 499 (directed study): Research on the financial and information aggregation properties of prediction markets (Fall 2024, Winter 2025)
- Satyam Goyal
  Project: Evaluating LLMs as arbiters in the fair allocation of indivisible items to agents with subjective valuations (Winter 2025)

- **Master's students supervised**
  - Arshdeep Singh (jointly with Michael Wellman)
    Project: An agent-based model of pathology propagation, detection, and mitigation in a financial market network (Fall 2022 – Winter 2024)
  - Tingfu Zhou
    Master of Data Science Capstone Project: EMOchecker: An Efficient and Cost-Friendly Affective Judgment Prompting Method for Multimodal LLMs (Fall 2024)

- **Faculty grader** for the 2024 cohort of the Multidisciplinary Design Program (MDP), College of Engineering, University of Michigan